Faculty of Engineering & Technology

Electrical & Computer Engineering Department

ENCS3340

ARTIFICIAL INTELLIGENCE

Project #2

Comparative Study of Image Classification Using Decision Tree, Naive Bayes, and Feedforward Neural Networks

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Table of Contents

Table of figure 3

Introduction 4

Dataset description 4

Models explanation 5

- **Naive Bayes model** 5
- **Decision Tree** 5
- **Feedforward Neural Network (MLP Classifier)** 5

Evaluation results with accuracy metrics and confusion matrices 6

- **Naive Bayes Classifier Results** 6
- **Decision Tree Classifier Results** 7

  Visualize and analyze decision paths 8
- **Feedforward Neural Network** 9

Comparative analysis and discussion 10

Conclusion 11

Link of google drive for dataset 11

Table of figure

Figure 1 : Naive Bayes Classifier Results 6
Figure 2 : decision Tree Classifier Results 7
Figure 3 : analyze decision tree path 8
Figure 4 : feedforward Neural Network 9

Introduction and dataset description

Introduction

Image classification is a common task in the field of artificial intelligence. In this project, we work on classifying grayscale images into three categories: birds, cats, and fish. The main goal is to build models that can recognize patterns in the images and predict the correct class for new ones. We use a labeled dataset and apply three different models: Decision Tree, Naive Bayes, and Multi-Layer Perceptron (MLP). After training the models, we evaluate them using accuracy scores and confusion matrices. Finally, we compare their results to understand which model works best.

Dataset description

The dataset used in this project contains images of three classes, birds, cats, and fish. All images are converted to grayscale to simplify processing and reduce computational. Each image is resized to a fixed dimension (32x32), flattened into a 1D vector, and then normalized before being fed into the models. The dataset is a folder that contains three subfolders cat with 170 images, fish with 186 and bird with 189 images so the total dataset is 545 images.

Models explanation

- **Naive Bayes model**

Naive Bayes is a simple model based on probability. It assumes that all features its pixels in our project are independent, which is not always true in images. it works surprisingly well in many situations. In our project, we use Gaussian Naive Bayes, which assumes the pixel values follow a normal distribution.

- **Decision Tree**

The Decision Tree model is one of the simplest classification methods. It splits the data into smaller groups by testing conditions on the pixels. In this project, each node checks the value of one pixel, and each branch shows the result of this test, leading to the final class prediction. Also, we used SelectKBest to pick the 200 most important pixels (features) to make the model faster and reduce noise.

And we applied Grid Search to try different settings, such as the type of splitting rule (gini or entropy), the maximum depth of the tree, and the minimum number of samples needed to split or create a leaf. This helped us find the best combination for higher accuracy.

Because Grid Search tests many parameter combinations and trains the model several times once for each combination, it takes more time to run compared to training the tree with just one set of parameters.

- **Feedforward Neural Network (MLP Classifier)**

The MLP (Multi-Layer Perceptron) is a type of neural network made up of layers of connected nodes called neurons. It can learn complex patterns and relationships in the data.

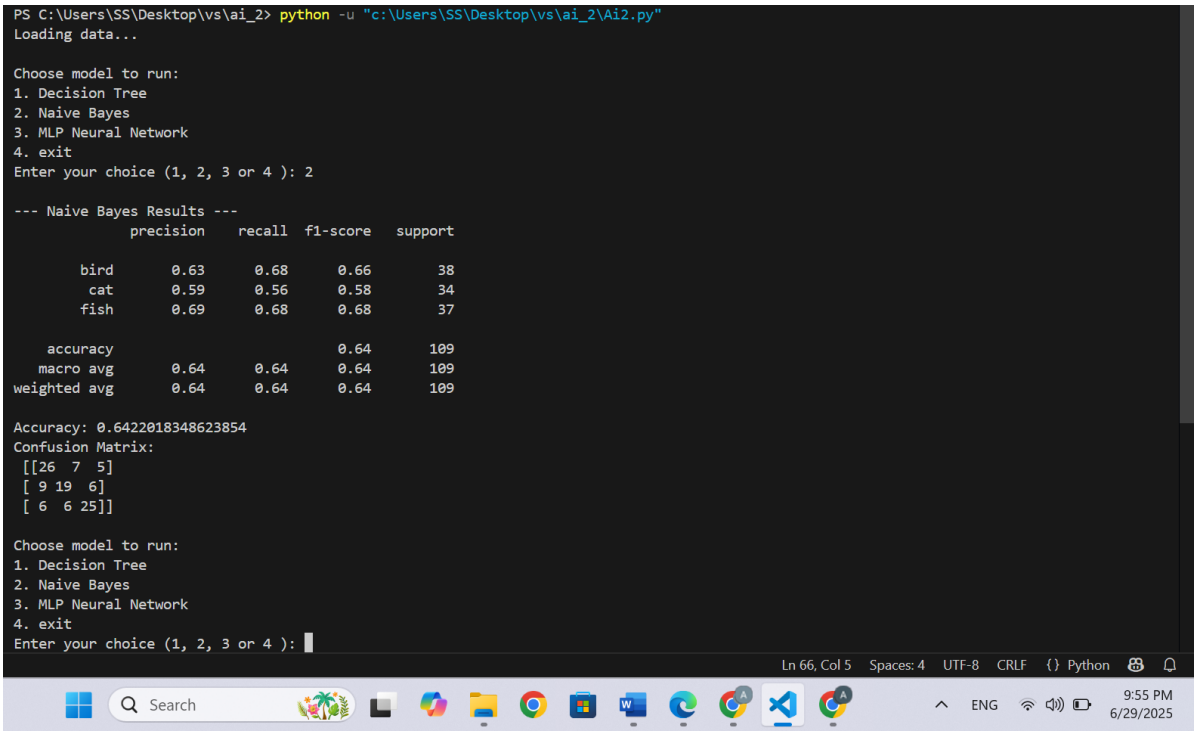
In our project, we used an MLP with two hidden layers the first layer has 100 neurons, and the second has 50 neurons. Having multiple hidden layers helps the network learn more detailed and abstract features from the images, which can improve accuracy compared to using only one layer.

We used the relu activation function to make learning faster and more effective, and the Adam optimizer to train the model efficiently. The learning rate was set to 0.001, and we allowed up to 2000 training iterations to give the model enough time to learn.

While MLPs are powerful and can achieve good results, they need more time and data to train compared to simpler models like Decision Trees or Naive Bayes.

Evaluation results with accuracy metrics and confusion matrices

- ### Naive Bayes Classifier Results



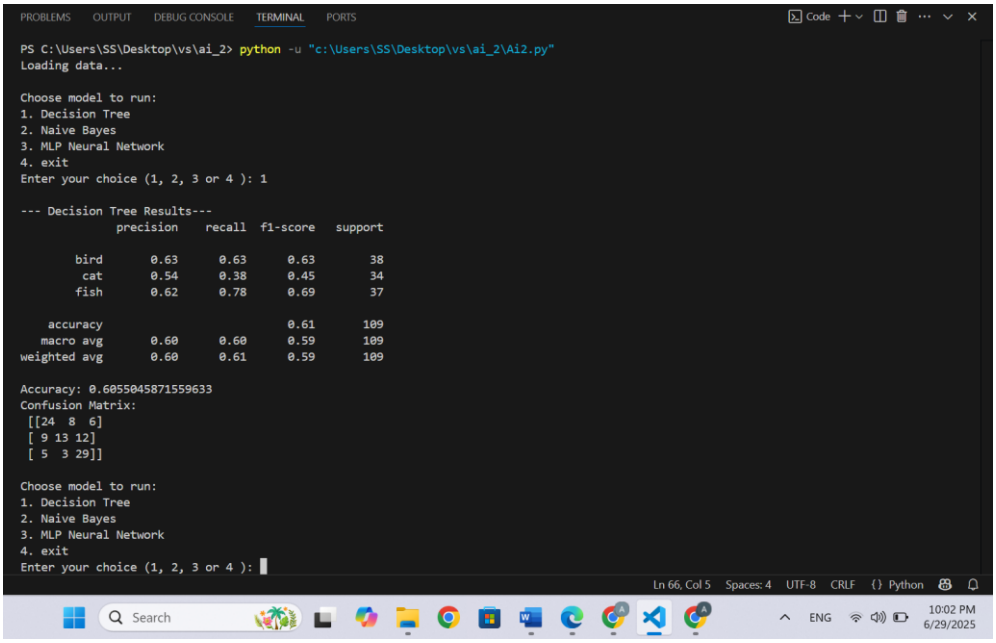
```
PS C:\Users\SS\Desktop\vs\ai_2> python -u "c:\Users\SS\Desktop\vs\ai_2\Ai2.py"
Loading data...

Choose model to run:
1. Decision Tree
2. Naive Bayes
3. MLP Neural Network
4. exit
Enter your choice (1, 2, 3 or 4 ): 2

--- Naive Bayes Results ---
              precision    recall  f1-score   support

        bird       0.63      0.68      0.66        38
         cat       0.59      0.56      0.58        34
        fish       0.69      0.68      0.68        37

    accuracy                           0.64       109
   macro avg       0.64      0.64      0.64       109
weighted avg       0.64      0.64      0.64       109

Accuracy: 0.6422018348623854
Confusion Matrix:
 [[26  7  5]
 [ 9 19  6]
 [ 6  6 25]]

Choose model to run:
1. Decision Tree
2. Naive Bayes
3. MLP Neural Network
4. exit
Enter your choice (1, 2, 3 or 4 ):
```

Figure 1 : Naive Bayes Classifier Results

The Naive Bayes model achieved an accuracy of 64.2%, which is considered a reasonable performance given the nature of the dataset and image size. The dataset includes 189 bird images, 186 fish images, and 140 cat images, with all images resized to 32x32 grayscale. Despite being a simple probabilistic model, Naive Bayes performed particularly well on the fish and bird classes, achieving precision and recall values above 0.63 for both.

The cat class, which has the fewest images 140 images, showed slightly lower performance with a precision of 0.59 and recall of 0.56. This could be due to the smaller sample size and possibly higher visual variability among cat images. Overall, the model handled the balanced dataset fairly well, and its simplicity makes it a solid baseline method for this classification task. The results demonstrate that Naive Bayes can produce acceptable accuracy on low-resolution image data without the need for complex training

- **Decision Tree Classifier Results**



```
PS C:\Users\SS\Desktop\vs\ai_2> python -u "c:\Users\SS\Desktop\vs\ai_2\Ai2.py"
Loading data...

Choose model to run:
1. Decision Tree
2. Naive Bayes
3. MLP Neural Network
4. exit
Enter your choice (1, 2, 3 or 4 ): 1

--- Decision Tree Results---
              precision    recall  f1-score   support

        bird       0.63      0.63      0.63        38
         cat       0.54      0.38      0.45        34
        fish       0.62      0.78      0.69        37

    accuracy                           0.61       109
   macro avg       0.60      0.60      0.59       109
weighted avg       0.60      0.61      0.59       109

Accuracy: 0.6055045871559633
Confusion Matrix:
 [[24  8  6]
 [ 9 13 12]
 [ 5  3 29]]

Choose model to run:
1. Decision Tree
2. Naive Bayes
3. MLP Neural Network
4. exit
Enter your choice (1, 2, 3 or 4 ):
```

Figure 2 : decision Tree Classifier Results

The results of the Decision Tree model show an accuracy of approximately 60.5%, which is considered acceptable given the simplicity of the model and the nature of the dataset. The classifier performed particularly well on the Fish category, achieving a high recall of 78%. This suggests that fish images in the dataset are visually distinct and easier for the model to recognize, possibly due to consistent shapes or background features. The Bird class also achieved balanced precision and recall which are both around 63%, indicating relatively stable performance. However, the model struggled with the Cat category, showing a lower recall of 38%, which means many cat images were misclassified. This is likely due to the high variability in cat images. Overall, these results are consistent with what can be expected from a Decision Tree classifier working on low resolution grayscale images.

Visualize and analyze decision paths

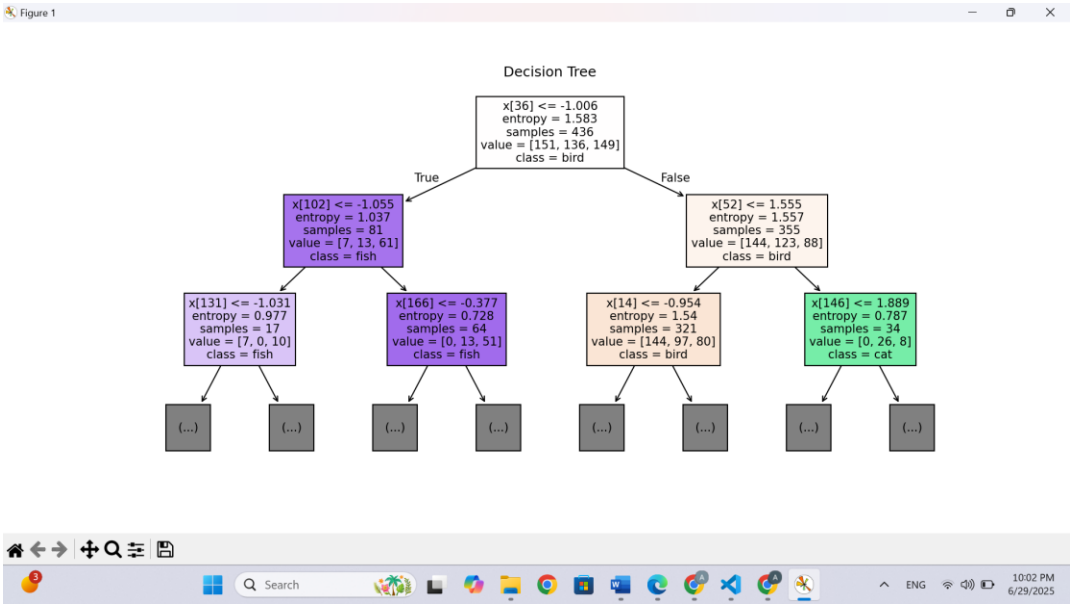
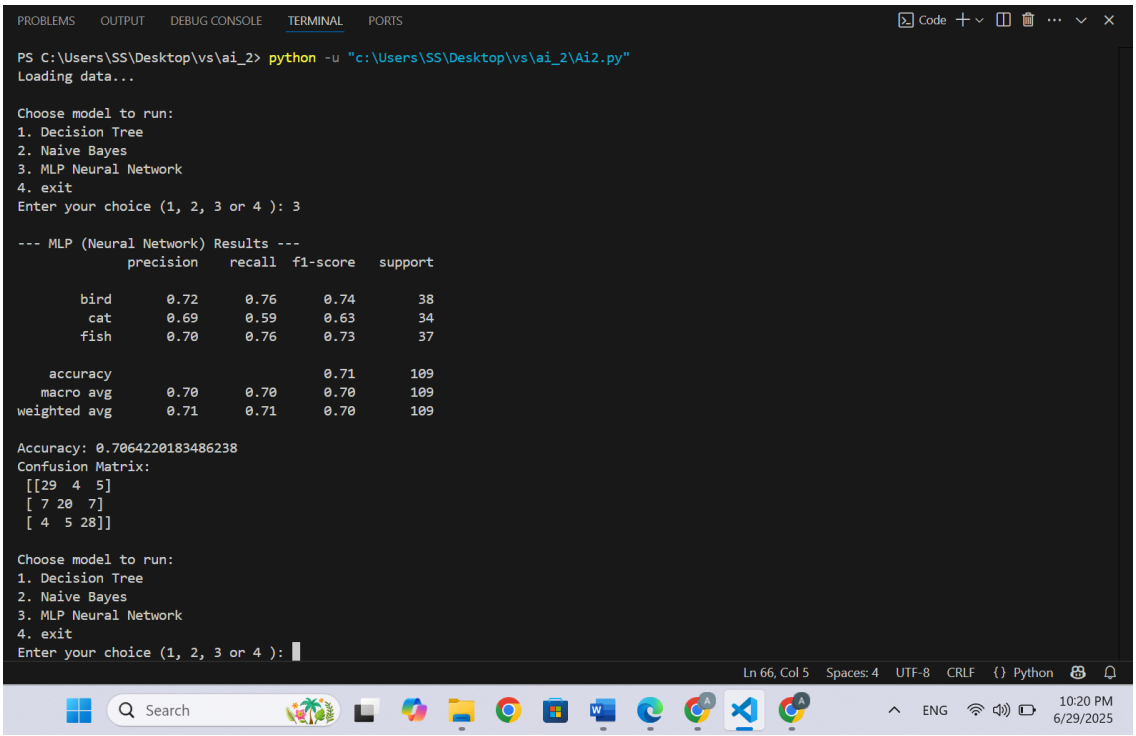


Figure 3 : analyze decision tree path

The decision tree illustrates the top layers of the trained classifier, showing how it splits image data based on pixel values. Each node represents a decision point, where the model checks whether a specific pixel is below or above a certain threshold. Based on the outcome which is True or False, the model routes the sample down the corresponding branch.

At the root node (x[36] <= -1.006) the model begins by dividing the data based on the intensity of pixel 36. If this condition is true, the sample tends toward the Fish class otherwise, it likely belongs to Bird or Cat. As we go deeper, the branches refine the classification. For instance, the path through x[102] <= -1.055 and then x[166] <= -0.377 leads to a leaf where 64 out of 64 samples are mostly is Fish. Similarly, on the right side of the tree, we see that node x[146] <= 1.889 isolates a subset where 26 out of 34 samples are Cat, suggesting that pixel 146 is a strong indicator for identifying cats in the dataset.

- ## Feedforward Neural Network



```
PS C:\Users\SS\Desktop\vs\ai_2> python -u "c:\Users\SS\Desktop\vs\ai_2\Ai2.py"
Loading data...

Choose model to run:
1. Decision Tree
2. Naive Bayes
3. MLP Neural Network
4. exit
Enter your choice (1, 2, 3 or 4 ): 3

--- MLP (Neural Network) Results ---
              precision    recall  f1-score   support

        bird       0.72      0.76      0.74        38
         cat       0.69      0.59      0.63        34
        fish       0.70      0.76      0.73        37

    accuracy                           0.71       109
   macro avg       0.70      0.70      0.70       109
weighted avg       0.71      0.71      0.70       109

Accuracy: 0.7064220183486238
Confusion Matrix:
 [[29  4  5]
 [ 7 20  7]
 [ 4  5 28]]

Choose model to run:
1. Decision Tree
2. Naive Bayes
3. MLP Neural Network
4. exit
Enter your choice (1, 2, 3 or 4 ):
```

Figure 4 : feedforward Neural Network

The MLP model achieved an accuracy of approximately 70.6%, which is a strong result for this classification task. The model performed consistently across all three classes birds, cats, and fish with precision and recall scores around 0.70. This balance indicates that the MLP was not only able to correctly identify the majority of samples, but also maintained a good trade off between false positives and false negatives. The confusion matrix further confirms that the model was especially accurate in predicting bird and fish images, while cat images had slightly more misclassifications.

This performance is considered good given the relatively small dataset size only 109 test images and the use of flattened grayscale images 32×32 resolution, which limits spatial feature learning. The model's ability to generalize well across the different classes despite these constraints suggests that the MLP is effectively learning complex patterns in the data.

Comparative analysis and discussion

After training and testing the three utilized models: Naive Bayes, Decision Tree, and Feedforward Neural Network (MLPClassifier), we performed an extensive comparison using different performance metrics including accuracy, precision, recall, F1-score, as well as confusion matrix analysis. Summary of the findings is shown below:

Model	Naive Bayes	Decision Tree	Feedforward Neural Network (MLP Classifier)
Accuracy	0.6422018348623854 =64.2%	0.6055045871559633 =60.5%	0.7064220183486238 =70.6%
Strengths	. Very fast at training and prediction. . Effective for data with flat pixel values. . Offers acceptable performance as a baseline, if the data is clean and the differences between classes are statistically significant.	. Easy to interpret and visualize. . Can learn non-linear rules from pixel intensities and can also handle interactions between features. . Improves when logical conclusions can be drawn based on some thresholds.	. Works very well in visually similar image classification. . Generalizes well if properly tuned (number of nodes and layers). . Able to learn complex nonlinear relationships and detailed feature representations.
Limitations	. He cant distinguish between complex explanations and non-shared boundaries between categories. . It assumes feature independence, which is often not the case in image data.	. Very sensitive to small changes in data . Prone to overfitting with noisy or small datasets.	. More difficult to interpret compared to tree based models. . Slow learning and takes time to adjust para meters.
Performance	. It performs relatively well but struggles to distinguish visually similar categories.	. Achieved average accuracy, at times outperfor ming Naive Bayes. . performance on test data decreased, indicating limitations in generalization.	. Reached the best values in accuracy, positive accuracy, and F1 for all classes. .Had a good ability for gener alization and dealing with un seen data.

Conclusion

In conclusion, although the Naive Bayes classifier provides a simple and fast baseline and the Decision Tree provides simplicity and understandable interpretation for image classification, the MLPClassifier neural network showcased undisputable supremacy in accuracy and generalization capability as well as effectiveness in recognizing subtle patterns in images.

Therefore, for critical image classification issues where accuracy matters the most, MLP is recommended if adequate resources are available. If execution speed is essential or computational power constraints exist, the Decision Tree model may be used as a viable substitute.

This comparative analysis emphasizes the need for selecting a model not just by crude accuracy, but also by interpretability, rapidity, and the type of data employed.

Link of google drive for dataset

https://drive.google.com/drive/folders/1A0FoKneR6JKrpCJdkDK7msOOyhmnjlXr?usp=sharing